SafeRoad AI+

AI-Powered Road Risk Detection & Emergency Response System

Hackathon: IIT Madras Road Safety Hackathon

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Tagline: Predicting road risks, preventing accidents, and saving lives using AI.

Problem Statement

Road accidents caused by potholes, damaged roads, poor visibility, and delayed emergency response continue to increase across urban and rural areas. Existing systems lack real-time monitoring, predictive analysis, and effective citizen participation. Authorities often receive complaints late, leading to unsafe road conditions and higher accident risks.

• Lack of AI-driven road condition monitoring

• Delayed accident emergency response

• No centralized road risk analysis platform

• Poor visibility of accident-prone zones

• Limited public participation in reporting road hazards

Proposed Solution

SafeRoad AI+ is an intelligent road safety ecosystem that combines AI-based road damage detection, risk score prediction, hazard mapping, and emergency SOS services into one platform. The system analyzes uploaded road images using computer vision models and assigns a dynamic road risk score. The platform helps citizens, authorities, and emergency responders identify dangerous roads proactively.

- User uploads road image using mobile application
- AI model detects potholes, cracks, and waterlogging
- Risk score is generated for the road segment
- GPS coordinates are stored automatically
- Hazard zones appear on live map
- Emergency SOS alerts are triggered during accidents

Core Features

AI Road Damage Detection

Detects potholes, cracks, and damaged roads using YOLOv8 computer vision models.

Road Risk Score

Assigns risk levels such as Safe, Moderate Risk, and High Risk based on road conditions.

Hazard Heatmap

Displays dangerous zones and accident-prone roads on a live map interface.

Emergency SOS

Provides one-click emergency alert system with GPS location sharing.

Authority Dashboard

Enables authorities to monitor complaints and prioritize repairs.

Technology Stack & Architecture

Component	Technology
Mobile App	Flutter
Backend	Firebase
AI/ML	YOLOv8 + OpenCV
Database	Firestore
Maps	Google Maps API
Dashboard	React.js

System Architecture:

Mobile App → AI Detection Engine → Risk Score Analysis → Firebase Database → Dashboard & Hazard Map → Emergency Services

Innovation & Social Impact

SafeRoad AI+ introduces predictive road risk analysis into conventional road safety systems. Instead of only reporting issues, the platform predicts dangerous roads and helps authorities take preventive action before accidents occur.

• Improves road safety awareness

• Reduces accident probability

• Enables faster emergency response

• Encourages citizen participation

• Supports future smart city infrastructure

Future Scope & Conclusion

Future Scope:

• IoT sensor integration for smart roads

• Tamil and multilingual voice reporting

• AI-based accident hotspot prediction

• Offline SOS alert via SMS

• Government smart city deployment

Conclusion:

SafetyRoad AI+ provides a scalable, intelligent, and practical solution for reducing road accidents through predictive AI monitoring and real-time emergency support. The proposed system can help create safer transportation infrastructure and improve public safety.

GitHub Link: SafetyRoad-AI-Plus